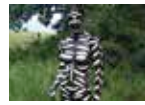




AI helps find unknown ancestors

Using deep learning methods on the Asian genome footprint of a new hominid who cross bred with its ancestors discovered. <https://digit.in/feb19-59>



Body-paint repels insects

A study shows that white, painted stripes on the body protect skin from insect bites. <https://digit.in/feb19-60>



WHAT IS A PLANET?

Considering how thoroughly the celestial bodies have been studied over the years, defining a planet should be a no-brainer. Turns out, there is a lack of scientific consensus on the issue.

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The Oxford English dictionary defines a planet as “A celestial body moving in an elliptical orbit round a star.” The word is Greek in origin, and the root means “wanderer”. The planets were distinguished from the stars, as they had specific motions

across the skies, independent of the stars. It is this motion that led to the discovery of planets beyond Saturn. In 1930, a young astronomer by the name of Clyde Tombaugh noticed a moving point of light in between two photographic slides in the direction of the constellation of Gemini. This slight movement indicated the discovery of a new planet, which was named Pluto.

In 1992, over a thousand trans-Neptunian objects (TNOs) were discovered beyond the orbit of Pluto. It looked

likely that there would be TNOs larger than Pluto. In 2003, one such object was discovered, with its own moon. It was the discovery of Eris, that led astronomers to first question exactly what a planet is. The media started referring to Eris as the tenth planet in the Solar System, which led to heated discussions within the International Astronomical Union (IAU). This is the body that is responsible for naming celestial objects such as stars, planets and asteroids.

Following an intense debate in the 2006 IAU general assembly, held at Prague, the definition of a planet was changed in a resolution. The resolution reads: (1) A planet is a celestial body that (a) is in orbit around the Sun, (b) has sufficient mass for its self-gravity to overcome rigid body forces so that it assumes a hydrostatic equilibrium (nearly round) shape, and (c) has cleared the neighbourhood around its orbit. (2) A “dwarf planet” is a celestial body that (a) is in orbit around the Sun, (b) has sufficient mass for its self-gravity to overcome rigid body forces so that it assumes a hydrostatic equilibrium (nearly round) shape, (c) has not cleared the neighbourhood around its orbit, and (d) is not a satellite. (3) All other objects, except satellites, orbiting the Sun shall be referred to collectively as “Small Solar System Bodies”. On 24 August 2006, at the closing ceremony of the general assembly, the resolutions were passed. This was the historic moment when Pluto was “demoted” as a planet, and the Solar System was considered to have 8 planets. According to the new definition of a planet, Pluto was now a “dwarf planet”. As of today, the IAU recognises five dwarf planets, Ceres, Pluto, Eris, Makemake and Haumea. Sedna, Orcus, Quaoar are some TNOs, that are not yet recognised as dwarf planets by the organisation.

Almost immediately, scientists began seeing some serious flaws in the IAU definition of the planet. The third criterion requires a planet to have cleared its orbit of other objects. This has a number of implications. None of